Database Systems

Lecture 1: Relational Databases

A relational database stores data in tables made of rows (records) and columns (attributes). A primary key uniquely identifies each row of a table, and a foreign key is a column that references the primary key of another table.

SQL (Structured Query Language) is the standard language for relational databases. SELECT reads data, INSERT adds rows, UPDATE modifies rows, and DELETE removes rows.

Lecture 2: Normalization

Normalization organizes tables to reduce redundancy and avoid update anomalies.

First Normal Form (1NF) requires that every column holds atomic values and there are no repeating groups. Second Normal Form (2NF) requires 1NF and that every non-key attribute depends on the whole primary key, not just part of it. Third Normal Form (3NF) requires 2NF and that there are no transitive dependencies, meaning non-key attributes depend only on the key.

Lecture 3: Transactions and ACID

A transaction is a sequence of operations executed as a single logical unit of work.

ACID stands for Atomicity, Consistency, Isolation and Durability. Atomicity means all operations of a transaction succeed or none do. Consistency means a transaction moves the database from one valid state to another. Isolation means concurrent transactions do not interfere with each other. Durability means once a transaction is committed it survives system failures.

An index is a data structure, usually a B-tree, that speeds up data retrieval at the cost of extra storage and slower writes.